

Study and analysis of a real complex network

XARXES COMPLEXES

Author: Maitane Fariñas Argoitia
Facultat de Física, Universitat de Barcelona

1 Introduction

Complex networks are powerful tools that allow us to model and analyse a broad variety of systems, from social networking sites to biological systems and technological infrastructure. By analysing a complex network, we can discover hidden patterns, identify clusters, uncover neighbour communities and even have an understanding of how information or diseases propagate.

These networks are made of interconnected nodes, where each of those nodes represents a different entity – a city, individual, social media profile...), and the connections between them, named edges, represent their interactions –roads, infection, friendship...).

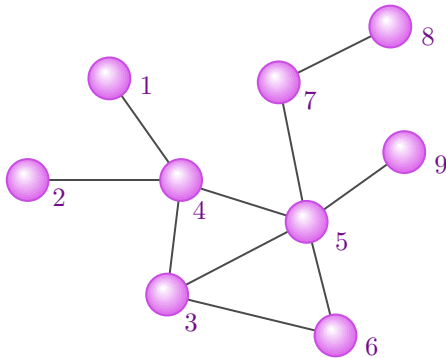


Figure 1: Example of a small complex network, with nodes in purple and edges in grey.

Real networks (such as the Internet or the plane connections of different cities across the world) exhibit some fascinating properties that distinguish them from regular networks and random networks, such as their high number of nodes or the structural heterogeneity. We will focus on these properties in the forthcoming sections.

In this work, we will focus on the study of a specific network, which will be introduced in the following section, by calculating some of its properties, such as the degree distribution, the average nearest neighbour's degree, and the clustering coefficient. These results will then be compared to the ones obtained from model networks. Finally, we will conduct a dynamic process, a simulation of an epidemic process, in which we will be able to see how a *disease*

¹Meaning that connections (i,j) and (j,i) are considered a single edge

spreads across the network in function of the infection- and recovery-rates.

2 Description of the network

The complex network chosen for the study is an email communication network that contains mailing activity of the Rovira i Virgili University, Tarragona. Each of the nodes represents a user, and the edges indicate that at least one email has been sent from one node to the other.

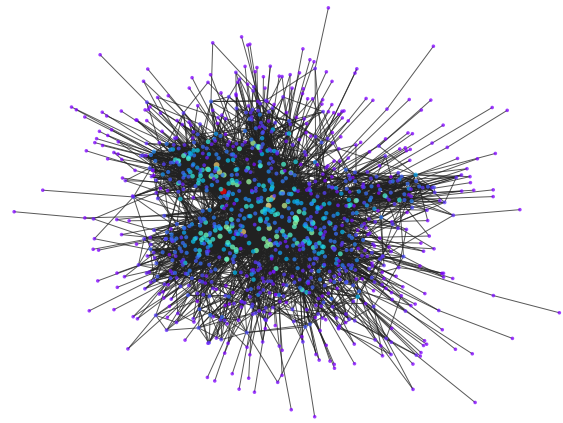


Figure 2: Picture of the chosen network.

It is a network constituted by $N = 1133$ nodes and $E = 5451$ non-repeating¹ edges. As previously stated, it is a communication network that considers connections as either sent or received emails, without making a difference; therefore, it is an undirected network. The way of measuring the interaction is simply checking whether there has been any communication between the users, not counting the number of exchanged mails, so it is an unweighted network, with no multiple edges. Also, no self-loop is allowed, given that the emails sent to oneself have not been taken into consideration.

Fig. 2 gives an idea of the shape and size of the network. Nodes' size and colour depend on the number of edges they

have: the more connections, the bigger the node.

3 Structural properties

This section will focus on the calculation of the structural properties of the chosen network. Before that, however, it may be of interest to describe how the storage of the network has been implemented.

3.1 Description of the network arrays

Most networks are introduced in the form of *edge-list*, in which all the edges of the network are listed by giving the indexes of the nodes that belong to them:

1	3
1	10
1	7
2	5
...	
i	j

All that information will be stored in the following three arrays:

- **Degree list:** $D_N = [k_1 \ k_2 \ \dots \ k_i \ \dots \ k_N]$.
The degrees of all nodes are listed such that k_i is in the i -th node's position.
- **Neighbour list:** $V_{2E} = \left[\overbrace{2 \ 4 \ 16 \ 17}^{k_1} \ \overbrace{1 \ 31 \ \dots \ i_N}^{k_2} \right]$.
The neighbours of all nodes are listed in order: the k_1 neighbours of node 1 in the first k_1 positions, the neighbours of node 2 from $k_1 + 1$ to $k_1 + k_2$ and so on.
- **Pointers:** $P_{N,2} = \left[[1, k_1], [k_1 + 1, k_1 + k_2], \dots \right]$.
This array stores the indexes of the first and last neighbour of each node, to make accessing the array V easier.

3.2 Calculation of the properties

Once the network is stored in an appropriate way, we can start calculating some of its properties.

Average degree

The average of the network is the average value of the number of edges per node, k_i :

$$\langle k \rangle = \frac{1}{N} \sum_i^N k_i \quad (1)$$

This value gives as a broad idea of the complexity and connectivity of the network: whether it is sparse or dense, and how well connected the nodes are on average.

The number of nodes N , the number of edges E and the average degree are connected in the following way

$$2E = N \langle k \rangle \quad (2)$$

The network we are studying, has an average degree of value $\langle k \rangle = 9.6222$. Real networks tend to be sparse, with a low value of N/E .

Degree distribution

The degree distribution is the probability distribution of the degrees over the whole network. It can be defined in different ways. The simplest one, the direct degree distribution, tells us the probability of a randomly chosen having degree k , $P(k)$. The direct degree distribution of our network can be seen in Fig. 3. The expression used to obtain such distribution is

$$P(k) = \frac{N_k}{N} \quad (3)$$

where N_k is the number of nodes of degree k , and N is the total number of nodes.

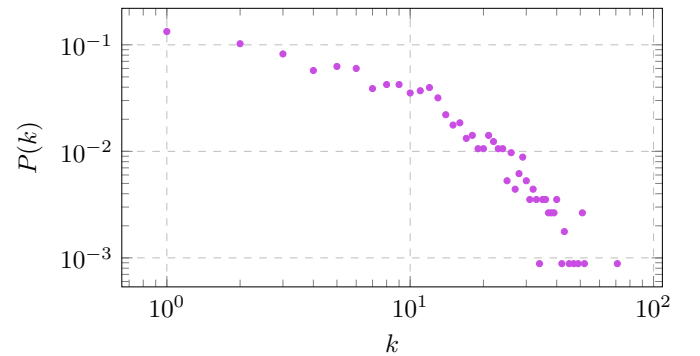


Figure 3: Direct degree distribution of the network.

Although Fig. 3 already shows a heterogeneous behaviour of the network and an overall idea of the distribution, the fluctuation or noise-level of the plot is rather high, which can make it difficult to distinguish possible underlying patterns.

In order to collect the most amount of information possible from the distribution, we will also calculate the cumulative complementary degree distribution, following the definition

$$P_c(k) = P(k_i > k) = 1 - \frac{1}{N} \sum_{k_i=1}^k n_{k_i} \quad (4)$$

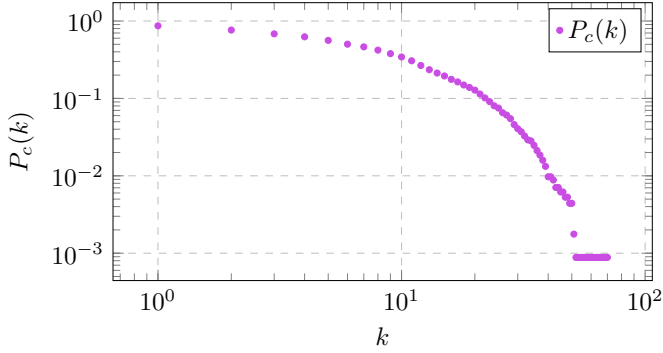


Figure 4: Cumulative complementary degree distribution of the network. Points represent the data of the network, and the dashed line is the fitting of the power-law.

In the the cumulative complementary distribution we've plotted in Fig. 4, we can see the heterogeneous behaviour of the distribution, with less fluctuations this time. Although with errors, the curve can be approximated, to some extent, with a power-law distribution, of the form of

$$P(k) \propto k^{-\gamma} \quad (5)$$

where γ is the exponent of the power-law, the parameter that reflects the degree of heterogeneity in the network. A smaller γ indicates a higher degree of heterogeneity, meaning that the network has a few highly connected nodes and lots of nodes with lower degrees: the $P_c(k)$ function decays slower, and, therefore, the nodes have a bigger range of k values. A bigger γ , therefore, suggests a more homogeneous network: the $P_c(k)$ decays fast, describing a network with a big number of nodes of same degree. Most real networks tend to have exponents in the range of $2 < \gamma < 3$, and have, as a result, a mathematically well defined mean over $k \in [1, \infty)$ and infinite variance. This kind of heterogeneous network is also called *scale-free* network.

In practice, few empirical phenomena obey power-laws for all values of k . In our network, the behaviour of $P_c(k)$ is quite different for small k than it is for the bigger k : the lower values display a more heterogeneous network (flatter), while at high degree values the function starts to decay faster, having different values of γ . The exponent of our network is $\gamma = 2.298299$ for the whole distribution, $\gamma = 1.507001$ for the lower values of k and $\gamma = 5.227165$ for the higher ones.

Average nearest neighbour degree

The average nearest neighbour degree (ANND) of a node of degree k is a measure of the dependencies between degrees of neighbour nodes in a network, or the tendency to connect to peers, in terms of the degree k . This property depends on the two-point correlations, encoded in $P(k', k)$, or, alternatively, $P(k'|k)$, the conditional probability. The latter describes the probability that an edge from a node with degree k connects to a node with degree k' . The ANND is a

more convenient way of portraying those correlations, as it helps to avoid fluctuations.

If the nodes that have many connections show a tendency to be connected to other nodes with many connections, the network is said to exhibit assortative mixing, which means that the average nearest neighbour degree will grow with the degree [1]. This behaviour is most common among social networks. On the other hand, if nodes with fewer connections tend to be linked to those with high degree, the network will have disassortative mixing (locally disassortative, in terms of immediate connectivity). Most networks of types other than social manifest disassortativity. Although negative slopes of the ANND suggest local disassortative behaviour, the assortativity coefficient of the network can still be positive, as the latter describes the behaviour of the network at a global level, not just locally.

The ANND of the network can be obtained via the following formula:

$$\bar{k}_{nn}(k) = \sum_{k'} k' P(k'|k) = \sum_{k'} k' \frac{E_{k'k}}{kN_k} \quad (6)$$

where $N_k = NP(k)$ is the number of nodes with degree k and $E_{k'k}$ is a symmetric matrix that contains the number of links between the subgroups of nodes with degrees k and k' .

Fig. 5 shows the ANND of our network in function of the degree, k .

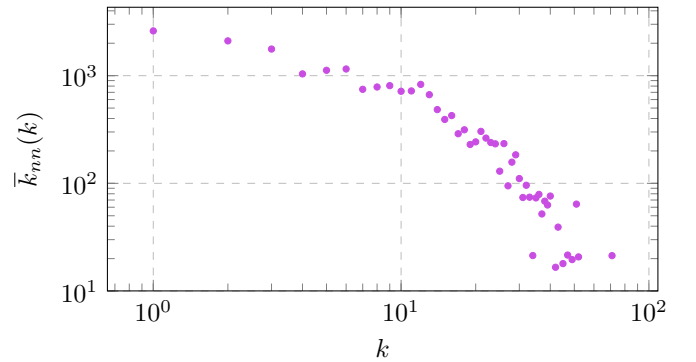


Figure 5: Average nearest neighbour degree of the network of disassortative behaviour.

We have labelled our network as a communications network, and, as most networks that do not fall into the social category, has disassortative mixing, which can be seen in Fig. 5: the value of $\bar{k}_{nn}$ decreases with k , which shows, as already stated, local disassortativity in terms of immediate connectivity. However, the assortativity coefficient is $\rho = 0.078201 > 0$, so the global behaviour of the network is assortative.

Clustering coefficients

The clustering of a network is a measure of the likelihood that two neighbours of a given node are neighbours themselves. The tendency of nodes to have many common neighbours is due to the correlations among three nodes, given by $P(k, k', k'')$ or $P(k''k'|k)$, which lead to the creation of triangles among neighbour nodes.

The clustering coefficient is the quantification of such property, and describes the probability that two neighbours of a node, chosen at random, are connected to each other. It is a measure of the number of triangles on a network. The degree dependent clustering coefficient, $\bar{c}(k)$, is defined as the number of triangles on a network divided by the total number of possible triangles, and averaged over degree classes:

$$\bar{c}(k) = \frac{1}{N_k} \sum_{i|k_i=k} \frac{2t_i}{k_i(k_i - 1)} \quad (7)$$

where t_i is the number of edges between the neighbours of node i , or, the number of triangles among node i and its neighbours.

It is also important to check whether the clustering of the network is weak or strong. If the triangles are disjoint, as in the network (a) of Fig. 6, the clustering is considered weak; instead, if the triangles coalesce, as in the network (b) of Fig. 6, it is considered to be strong clustering. The limit between the two is given by the function

$$\bar{c}(k) = \frac{1}{k - 1} \quad (8)$$

It is considered weak clustering if $\bar{c}(k) < 1/(k - 1)$

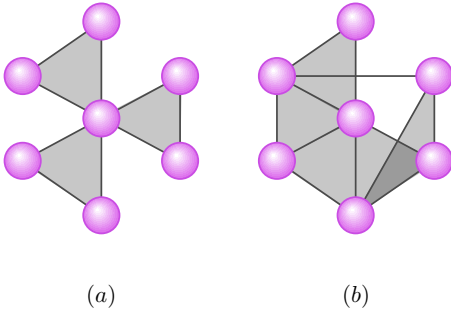


Figure 6: Weak clustering, (a) vs strong clustering, (b).

The degree dependent clustering coefficient of our network is displayed in Fig. 7.

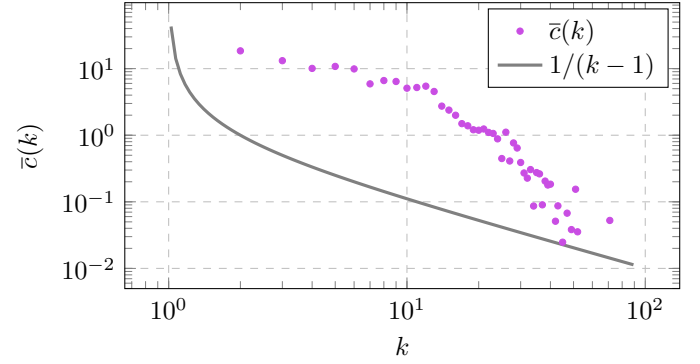


Figure 7: Degree depending local clustering coefficient. The data from the network in purple, and the limit between strong and weak clustering in grey.

It is quite clear that our network exhibits strong clustering, since all nodes have a bigger value of $\bar{c}$ than the one given by $1/(k - 1)$. That means the triangles are interconnected and, thus, the locally tree like assumption cannot be considered in this case.

To conclude with this section, we will analyse the global clustering coefficient. This parameter is defined as the total number of triangles in the network, divided by the number of connected triples:

$$C = \frac{3T}{\# \text{ of connected triples}} = \frac{3T}{\sum_k N_k k(k - 1)/2} \quad (9)$$

where T corresponds to the number of triangles of the network, without counting any of them twice. The total number of triangles in our network is $T = 5343$ (or 16029, if we count triangles as the sum of the triangles for each node), the number of connected triples is 96415, and from that, the corresponding value of global clustering of our network is $C = 0.166250065$.

Average path length

The average path length of a network is the average of the shortest path length between nodes i and j . The shortest path length is defined as the smallest number of steps needed to get from node i to node j .

There are several algorithms to implement this property, each; in this case, the Dijkstra's algorithm has been employed.

Since it is a very computationally-expensive property to calculate, we have just implemented it in the following section, in 4.1, for smaller networks.

4 Network models

Network models are mathematical tools that allow us to study the effects that certain properties of the real complex networks may have on the wider behaviour of the system.

The small world property, heterogeneous degree distribution, degree correlations and high levels of clustering are some of the features that are repeatedly found in real networks, and building network models is one of the best way to understand the emergence of such characteristics.

In the forthcoming section, we will focus on two of the equilibrium models, the small world model and the configuration model. The aim of this section is to create different networks, each of them with a different set of constraints (number of nodes N and average degree $\langle k \rangle$, or degree list D_N), and either check whether they are able to replicate real network's features, or, given the values of our network, to compare the properties of the resulting networks to our network's.

4.1 Small World model

The small world model (SW) constructs a complex network given the number of nodes N , the average degree $\langle k \rangle$ and a parameter β . It is an intermediate case between the regular network, where all nodes have the same degree $k_i = k = \langle k \rangle$, and a structure close to the the Erdős-Rényi $G(n, p)$ random model with $p = \langle k \rangle / (N - 1)$, where connections between nodes are established randomly, connecting each pair of nodes with probability p . It is possible to go from one model to the other by varying the parameter β , the rewiring probability, being $E_r = \beta N \langle k \rangle / 2$ the number of non-lattice edges introduced: $\beta = 0$ for the regular model, $\beta = 1$ for the random one, and values in the range $0 < \beta < 1$ for the SW model.

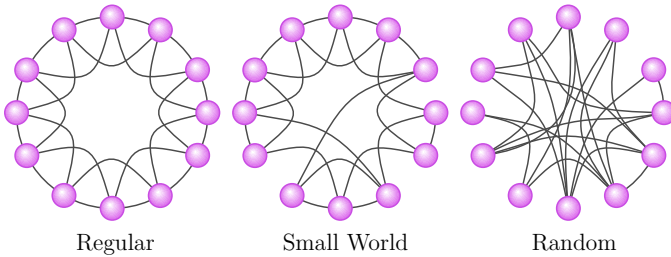


Figure 8: Pictures of a regular network model, with $\beta = 0$; a SW model; and a random network model.

Regular networks are known for having high clustering; however, they also tend to have high average path lengths (not small world); on the other hand, random networks do not produce clustering (they can have large clustering coefficients due to the finite size effects), but are able to reproduce the small world property. In the SW model, both the small world property and high values of clustering can be found, which are two of the features of real complex networks.

We will build the SW model with the Watts-Strogatz mechanism, and plot the values of the global clustering coefficient, C , and average path length, $\bar{l}$, in function of the

parameter β .

First, it might be convenient to check that the parameters we get from the model we created match the expected values. In order to do so, we will compare the global clustering coefficient for $\beta = 0$ and $\beta = 1$ with the analytical values of the regular and random models, respectively.

Regular model

We know that the value of the global clustering coefficient is given by Eq. (9). Since $\beta = 0$ corresponds to the SW limit of the regular model, with zero randomness, we can get the value of C by either counting the triangles analytically, or with the method of calculating the clustering coefficient, as in the previous section.

In order to obtain C analytically, we need to write an expression for $t_i(k)$, the number of triangles per node, in function of the degree:

$$t_i(\langle k \rangle) = \frac{1}{2} \sum_{k \text{ even}}^{\langle k \rangle} (k - 2) = \frac{\langle k \rangle^2 - 2\langle k \rangle}{8} \quad (10)$$

Since it is a regular network, all nodes will have the same degree, therefore $k_i = \langle k \rangle$; the number of nodes of degree k will be $N_k = N$ if $k = \langle k \rangle$, and $N_k = 0$ otherwise. Replacing these values in Eq. (9), we get the following formula for the global clustering coefficient:

$$C(\langle k \rangle) = \frac{6t_i}{\langle k \rangle (\langle k \rangle - 1)} = \frac{3}{4} \frac{\langle k \rangle - 2}{\langle k \rangle - 1} \quad (11)$$

For the parameters of network [3], $\langle k \rangle \approx 10$ and $N = 1133$, our SW model, with $\beta = 0$, gives us a global clustering coefficient of $C = 0.666666$, which is exactly the same as the one obtained from Eq. (11), with $t_i = 10$ triangles per node.

The global clustering coefficient of the regular network is almost 4 times bigger than the one we get in our real network, with the same number of nodes and practically equal average degree.

Random model

Now we will repeat the same procedure for the random model, by setting $\beta = 1$ in our SW model.

In a random network, the global clustering coefficient is given by

$$C = p = \frac{\langle k \rangle}{N - 1} \quad (12)$$

which for $N = 1133$ and $\langle k \rangle = 10$ gives a value of $C = 0.00883$. From the SW model, with $\beta = 1.0$, we obtain values around $C = 0.007864$, which is somewhat close to the previous one.

Now that we have made sure the values of C are correct for limit cases of the SW model, we can create different networks by varying β ; in this manner, we can identify the optimal values of β that yield the desired outcome: high values of C combined with low average shortest path length, L .

Finally, we will calculate how the global clustering coefficient, C , and the average shortest path, $\bar{l}$, vary depending on the value of β , to see that the SW model indeed replicates the properties of small world and short average path.

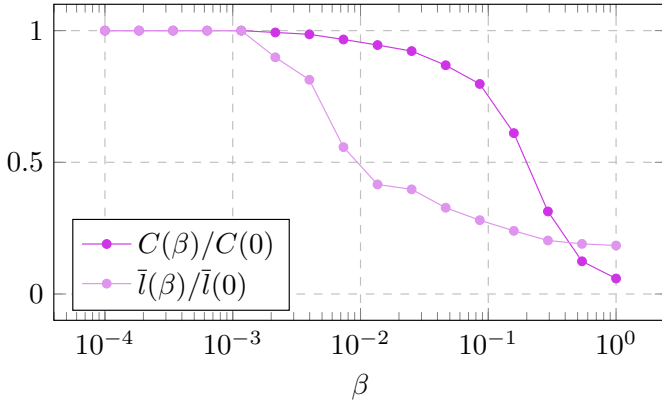


Figure 9: Clustering coefficients and average shortest paths for different β values of the SW model. $C(0)$ and $\bar{l}(0)$ correspond to the values of the regular network of $N = 200$, $\langle k \rangle = 6$.

Fig 9 shows the obtained values for networks² with $N = 200$ nodes and $\langle k \rangle = 6$.

Fig. 9 shows that, as we previously stated, there is a region of β in which both the small world property and small average shortest path conditions are met. Values of $0.01 \lesssim \beta \lesssim 0.2$ display ratios of clustering coefficients with $C/C(0) > 0.5$, while of the average shortest path is $\bar{l}/\bar{l}(0) < 0.5$.

4.2 Configuration model

The configuration model is another method for generating random network, but, in this case, the degree sequence –degree list, D_N – is the constraint. It is widely used as a reference model for real complex networks, because it allows the modeller to incorporate arbitrary degree distributions. It produces maximally random networks for the preassigned degree sequence, uncorrelated except for structural unavoidable correlations necessary to close the networks.

In this model, a sequence of *half edges* –or stubs– is assigned to the nodes, following the given degree sequence, and those stubs are then connected randomly, each of the

²These parameters have been chosen instead of the ones in our network, because calculating $\bar{l}$ involves computationally expensive operations. Performing this calculation multiple times with a network of this size would have taken an unreasonably long time.

matching possibilities having the same probability. Therefore, self loops or multiple edges are also accepted, if not manually rejected. Biases are introduced at this point, since those self loops and multi-edges must be avoided and, in order to do so, we must follow a non-uniform sampling from the given ensemble (by rejecting some connections).

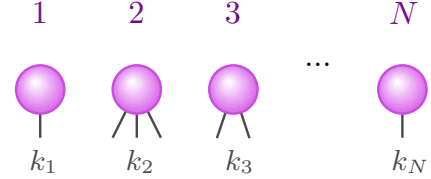


Figure 10: All nodes of the network listed along their corresponding number k_i of half edges or stubs.

So, in this case, the stubs are the ones picked with equal probability, and, therefore, nodes with high degree will have more probability of being chosen for a new connection.

By comparing the results of obtained from the CM with the ones of our real network, we will see whether this model is able to replicate the ANND and the degree dependent clustering coefficient based on the degree distribution.

Fig. 11 and 12 show the ANND and the degree dependent clustering coefficient of both the real and the CM networks, respectively.

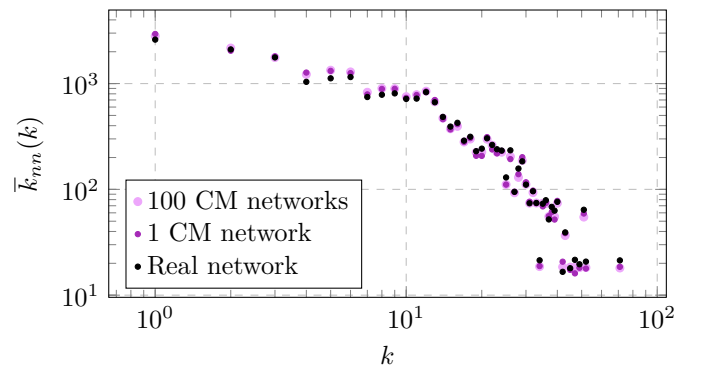


Figure 11: Average nearest neighbours degree.

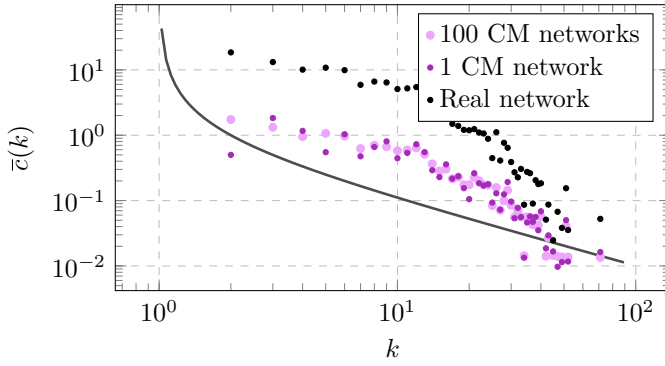


Figure 12: Degree dependent clustering coefficient.

From Fig. 11, we can conclude, since the average nearest neighbours of both networks coincide, that the ANND property is mainly dependent on the degree distribution of a network, as that is the only constraint introduced in the CM. On the other hand, Fig. 12 shows a bigger disparity between both models: the clustering coefficient of the real network is a whole order of magnitude bigger than the one from the CM, and the latter shows weak clustering for high values of k .

This means that the degree dependent clustering coefficient does not only depend on the degree distribution of the network; there are more characteristics to take into account: the local structure of the network, the degree of each node...

5 Dynamical process

To conclude with the project, we will study and simulate a dynamical process in the network.

The ultimate goal in studying networks is to better understand the behaviour of the systems networks represent [2]. The analysis of the structural properties and network models, such as the ones discussed in sections 3 and 4, take us a step closer to reaching that goal, but there is still the need to turn those results into predictions about how the overall system will behave.

In this section we will conduct a dynamical process with the network we described and studied in the previous section [3]. We will run a simulation of an epidemic process, where, starting from a small portion of the population being infected, the evolution and spread of the infection will be measured. The model we will employ is the SIS model, which will be described in detail in the following section.

The first part will be focused on the analytical resolution of the problem, which is based on the integration of the differential equation of the prevalence of the system. On the other hand, in the last part will be immersed in the simulation of the epidemic process, based on the Gillespie algorithm.

5.1 SIS model

The susceptible-infected-susceptible (SIS) model is the simplest epidemic model that allows reinfection. There are just two states: susceptible (S) and infected (I); susceptible individuals (nodes) get infected, and those infected go back to the susceptible state after the recovery. This model is mainly used as a paradigmatic model for the study of infectious diseases leading to an endemic state with a stationary and constant value for the prevalence of infected individuals, i.e. the degree to which the infection is widespread in the population as measured by the density of those infected, which, in the following sections, will be denoted as ρ .

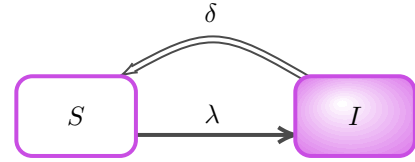


Figure 13: Flowchart of the SIS model; S for susceptible and I for infected. λ is the infection rate and δ the recovery rate.

5.2 Analytical model

The state of each node will be given by

$$n_i(t) = \begin{cases} 0 & \text{if susceptible,} \\ 1 & \text{if infected.} \end{cases} \quad (13)$$

Each susceptible node has a probability of getting infected, which depends on the number of infected neighbours and λ , the infection-rate. On the other hand, an infected node has a probability of recovery given by δ , the recovery rate. The evolution of $n_i(t)$ can be written as follows:

$$n_i(t + dt) = n_i(t)\xi_i(dt) + (1 - n_i(t))\eta_i(dt) \quad (14)$$

where

$$\eta_i(t) = \begin{cases} 1, & \text{with } p = \lambda dt \sum_j a_{ij} n_j(t), \\ 0 & \text{with } p = 1 - \lambda dt \sum_j a_{ij} n_j(t). \end{cases} \quad (15)$$

and

$$\xi_i(t) = \begin{cases} 0, & \text{with } p = \delta dt \\ 1, & \text{with } p = 1 - \delta dt. \end{cases} \quad (16)$$

The prevalence of the system is the parameter that describes the portion of infected population over time, $\rho_i(t) \equiv \langle n_i(t) \rangle$ and its differential equation can be obtained as:

$$\frac{d\rho_i}{dt} = -\delta\rho_i(t) + \lambda(1 - \rho_i(t)) \sum_{j \in V_i} \rho_j(t) \quad (17)$$

where we denoted the neighbours of node i as V_i . The pair approximation has been applied: $\rho_{ij} = \langle n_i(t)n_j(t) \rangle \approx \langle n_i(t) \rangle \langle n_j(t) \rangle = \rho_i \rho_j$.

The differential equation has been integrated with the Runge-Kutta method of fourth order. Fig. 15 shows the results obtained for our network, where the initial state has been chosen randomly, with an initial probability being infected of $p = 20/N$. The equation has been integrated up to $t = 15$, where the prevalence is in an equilibrium state.

Fig. 14 shows the time-evolution of the prevalence $\rho = \langle \rho_i \rangle$.

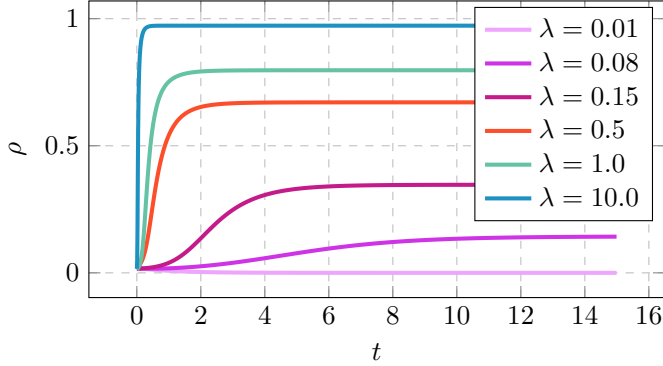


Figure 14: Time-evolution of the prevalence.

We can see that the time from $t = 0$ to equilibrium is bigger for smaller λ values, but they all end up reaching equilibrium before $t = 15$.

Fig. 15 shows the result of the prevalence ρ of the network in function of the infection-rate, λ . The recovery-rate has been set to $\delta = 1$, since it is the ratio λ/δ which defines the evolution. The mean-field approximation of the prevalence is also plotted, where $\rho = \langle \rho_k \rangle$ is the average of the prevalence for different values of degree, k .

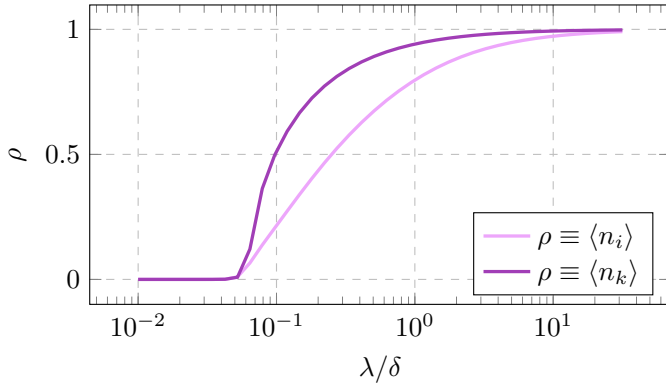


Figure 15: Prevalence as a function of the relative infection rate, λ/δ , integrated with the Runge-Kutta of 4th order.

Fig. 15 shows the prevalence in equilibrium in function of λ . The prevalence of the mean-field shows higher values of infection than the one without the approximation, reaching higher infected population portions faster. But the critical infection-rate is equal for both of them, around $\lambda_c = 0.05$.

5.3 Simulation

Epidemic processes can be defined as Markovian processes: the network has no memory and it does not take into account the previous history, the state of time $t + dt$ is only defined by the state of time t .

The evolution of the system is defined by the rate of events (recovery/infection): the higher the event rate, the shorter the time between the current event and the next one. Such time between events is denoted as τ , and it is a random variable, sampled from an exponential distribution of the form of:

$$\Psi(\tau) = \lambda_0 e^{-\lambda_0 \tau} \quad (18)$$

where τ is the total event-rate, given by:

$$\lambda_0 = N_I \delta + N_A \lambda \quad (19)$$

where N_I is the number of infected nodes, and N_A the number of active edges, edges with an active node in one end, and a susceptible in the other. For each event (*iteration*), a node is chosen randomly, with all infected nodes having the same recovery probability, and susceptible nodes with infection-probability proportional to the number of infected nodes.

In the simulation time is not defined by any computer time-step, it is τ the parameter that describes the evolution. The simulations we have carried out stop once the time has reached $t = 15$.

Fig. 16 shows the time-evolution of the prevalence obtained from the simulations, compared to the analytical ones.

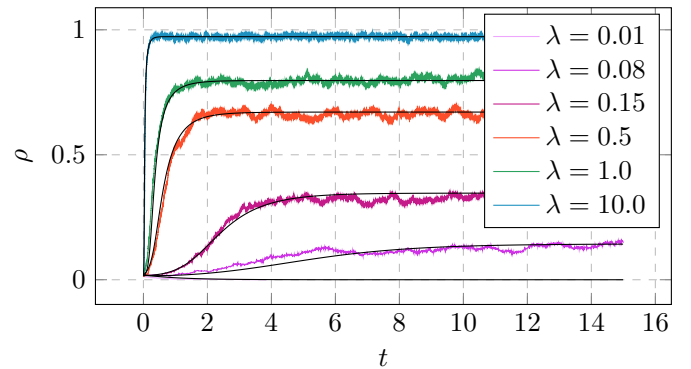


Figure 16: Time-evolution of the prevalence obtained from the simulations, compared to the analytical results.

Fig. 17 shows the results obtained from the simulation, compared to the ones from the analytical results. The higher the value of λ , the better the fit.

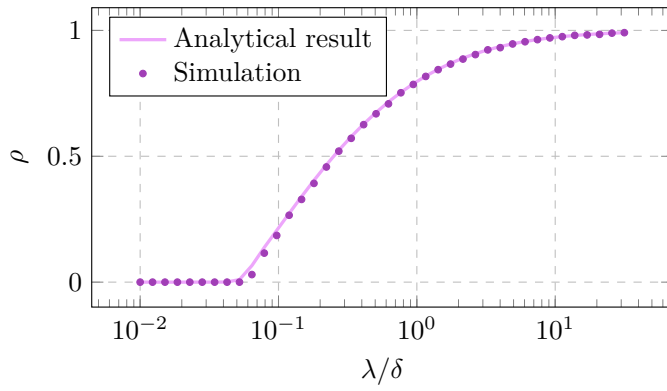


Figure 17: ρ as a function of the relative infection rate, λ/δ , obtained from the simulation.

Both results show exactly the same behaviour, the same growth for $\lambda \geq \lambda_c$, and almost identical value of λ_c .

6 Conclusions

In this project we have studied the structural properties of a network that contains the mail communications of the Rovira i Virgili University.

We have analysed several properties, and then compared them to ones obtained from two different network models: the Small World (SW) and the Configuration Model (CM). The results show that the SW model, for values of β around $0.01 \leq \beta \leq 0.2$. On the other hand, the CM shows average neighbour degrees compatible to the ones of our network,

but the clustering coefficient does not, since the latter does not only depend on the degree list of the network, but the local properties as well.

In the last section, we have carried out an epidemic process simulation. We integrated the analytical equations to compare the results with the simulations of Markovian process simulations, and both of the cases show almost identical time evolution. The system goes through a phase-transition depending the values of λ : when λ exceeds $\lambda_c \approx 0.05$, the small portion of initial infected population is capable of infecting a macroscopic portion of the system, reaching the epidemic phase.

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